SEMINAR

Types of Hazards

Biological • Mechanical • Physical • Electrical • Fire • Dust

Presented by: Sundus

Outline

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Introduction

What is a hazard?

A hazard is anything at work or in daily life that has the potential to cause injury, illness, or damage.

Why classify hazards?

Knowing the type of hazard helps us recognise it early and choose the right control measure.

Hazard

A source of potential harm

Risk

Chance of harm × how severe it is

Aim

Identify, assess and control hazards

1

Biological

HAZARDS

Living organisms or their products that can cause infection, allergy or disease. They can spread through air, water, food, or contact with infected people, animals or surfaces.

Examples

- Bacteria, viruses, fungi and parasites
- Blood and body fluids
- Animal bites, insects, mould

Control measures

- Hand washing and good hygiene
- Vaccination
- Gloves, masks and safe waste disposal

2

Mechanical

HAZARDS

Injuries caused by moving parts of machines, tools and equipment. Most accidents happen when machines are unguarded, poorly maintained, or used without proper training.

Examples

- Cutting, crushing and entanglement
- Flying or falling objects
- Unguarded rotating parts

Control measures

- Machine guards and safety devices
- Lockout / tagout during repair
- Training and regular maintenance

3

Physical

HAZARDS

Environmental factors that can harm the body without direct contact with a substance. These hazards are often invisible, and the harm usually builds up slowly with long exposure.

Examples

- Excessive noise and vibration
- Extreme heat or cold
- Radiation, poor lighting, high pressure

Control measures

- Ear protection and rest breaks
- Proper ventilation and lighting
- Shielding and limiting exposure time

4

Electrical

HAZARDS

Danger from contact with live electricity, causing shock, burns or fatal injury. Even a small current can be dangerous, so electricity must always be handled with care.

Examples

- Faulty or exposed wiring
- Overloaded sockets and circuits
- Working with wet hands or in damp areas

Control measures

- Insulation and proper earthing
- Circuit breakers and RCDs
- Only trained persons repair equipment

5

Fire

HAZARDS

Uncontrolled burning that can cause burns, smoke inhalation and property loss. A fire needs fuel, heat and oxygen to start, and removing any one of them can stop it.

Examples

- Flammable liquids and gases
- Short circuits and overheated equipment
- Smoking and open flames

Control measures

- Fire extinguishers and alarms
- Clear exits and evacuation plan
- Safe storage and regular fire drills

6

Dust

HAZARDS

Fine airborne particles that can be inhaled and may also cause explosions. Long-term breathing of dust can damage the lungs, and dust clouds in closed spaces can ignite.

Examples

- Silica, cement, wood and coal dust
- Flour and grain dust
- Metal powders and chemical dust

Control measures

- Ventilation and dust extraction
- Wet methods and good housekeeping
- Respirators / dust masks

Conclusion

1

Hazards are present everywhere: at work, at home and in the environment.

2

There are six main types: biological, mechanical, physical, electrical, fire and dust.

3

Every type of hazard needs its own suitable control measures.

4

Awareness, training and safe habits help prevent accidents.

Thank You

Any questions?

Sundus

